W4: COSY INFINITY Cross-Validation of FELsim Beamline Optimisation

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1 Objective

Cross-validate FELsim's 11-stage sequential Nelder-Mead beamline optimisation against COSY INFINITY's internal FIT command. Both codes solve the same problem: matching Twiss parameters at the undulator entrance of the UH MkV FEL transport line (118 elements, 23 quadrupoles) to the targets of Weinberg, Fisher & Li [1].

The goal is independent confirmation that the optimised quadrupole settings achieve the required undulator matching — not necessarily identical currents, but equivalent final Twiss.

2 Method

2.1 Translation to COSY FIT

FELsim's 11-stage optimisation was translated into 11 sequential COSY FIT blocks within a single FOX input file. Each FIT block:

- Declares variables (quadrupole currents) and bounds,
- Re-executes the full lattice (via `LATTICE`),
- Evaluates a combined MSE objective from the Twiss/dispersion goals,
- Iterates until convergence ($\varepsilon = 10^{-8}$, $N_{\max} = 1000$).

Optimised variables persist between FIT blocks, so each subsequent stage inherits the results of all prior stages.

2.2 Key Implementation Details

Combined MSE objective. COSY's FIT with individual objectives (`ENDFIT` with multiple `OBJ`) resets to initial values when *any* objective fails to converge. A single combined MSE objective avoids this:

$$\mathrm{OBJ}(1) = \frac{1}{N}\sum_{i=1}^{N} w_i\,(f_i - g_i)^2, \tag{1}$$

where f_i is the computed Twiss parameter, g_i the target, and w_i the weight. The sum is evaluated without `CONS()`, preserving DA derivatives for gradient-based optimisation (FIT algorithm 1).

Initial Twiss from beam distribution. FELsim computes Twiss from particle statistics of a Gaussian beam ($\sigma_x = 0.8\,\text{mm}$, $\varepsilon = 0.101\,\pi\,\text{mm mrad}$). The initial beta function is $\beta_0 = \sigma_x^2/\varepsilon \approx 6.34\,\text{m}$, not $\beta_0 = 1\,\text{m}$. Using the wrong initial Twiss gives completely incorrect transport ($\alpha_x = -2.8$ instead of ≈ 0 after Stage 1). This was the most critical implementation finding.

Dipole model. Hard-edge dipoles (`DIL` without `FC`) with edge angles from the beamline specification. The `DIL` β parameter is set to zero (no fringe field integral), meaning edge kicks use $\tan(\epsilon)/\rho$ without the fringe field correction term. See Section 4 for implications.

3 Results

3.1 Beam and Target Parameters

Table 1: Beam parameters and undulator Twiss targets.

Parameter	Value
Energy	40 MeV
Normalised emittance ε_n	$8\,\pi$ mm mrad
Geometric emittance ε	$0.101\,\pi$ mm mrad
Initial beam size $\sigma_{x,y}$	0.8 mm
Initial β_0	6.34 m
Target β_x	1.400 m
Target α_x	0.470
Target β_y	0.242 m
Target α_y	0.000

These parameters are identical for the S1 (2 ps) and S3 (0.5 ps) configurations — the transverse beam parameters and undulator Twiss targets do not depend on bunch length, so the optimisation result is the same for both.

3.2 COSY FIT Convergence

The full 11-stage FIT converges in a single COSY execution. Table 2 compares the final Twiss at the undulator entrance.

Table 2: Final Twiss at undulator entrance: COSY FR 0, COSY FR 1, FELsim, and targets.

Parameter	Target	COSY FR 0	COSY FR 1	FELsim NM
β_x (m)	1.400	1.400	1.400	1.400
α_x	0.470	0.470	0.470	0.470
β_y (m)	0.242	0.242	0.242	0.242
α_y	0.000	0.000	0.000	0.000
MSE	—	4.50e-9	7.93e-8	2.89e-5

Both COSY runs match the targets to high precision. COSY's gradient-based FIT (algorithm 1, using DA derivatives) converges $\sim 10^4\times$ more precisely than Nelder-Mead. FR 0 and FR 1 achieve comparable MSE ($\sim 10^{-8}$–10^{-9}), confirming that fringe fields do not degrade convergence when warm-starting is used.

3.3 Quadrupole Current Comparison

A detailed comparison of optimised quadrupole currents across FELsim, COSY FR 0, and COSY FR 1 is given in Table 4 (Section 3.6).

Stages 1–2 and 6–9 show excellent agreement across all three runs ($\Delta < 0.1$ A). The principal discrepancy is Stage 5 (double triplet, elements 33–43): COSY finds a local minimum with *negative* currents for elements 35 and 37, while FELsim (constrained to $[0, 10]$ A) finds all-positive currents. A four-point multistart over Stage 5 starting values confirmed that COSY's gradient-based optimiser consistently converges to the negative-polarity basin.

Stages 10 and 11 also differ from FELsim, partly because the different Stage 5 outcome propagates through the downstream lattice, and partly due to the dipole model difference discussed in Section 4.

3.4 Dispersion

The COSY-optimised solution shows residual horizontal dispersion $\eta_x \approx 33$ mm at the undulator entrance. The Stage 11 objective includes dispersion zeroing at element 92 (5th chromaticity quad) with weight 0.5, but does not constrain η'_x. Between element 92 and the undulator (element 117), the non-zero η'_x causes dispersion to accumulate through the final triplet drifts.

FELsim shows comparable residual dispersion at element 92 (0.017 in optimizer units), confirming this is a lattice design feature rather than a code discrepancy.

3.5 Single-Dipole Transfer Matrix Comparison

To quantify the dipole model difference, we compare the 6×6 transfer matrix of individual DPW-DPH-DPW "sandwiches" between FELsim and COSY under three fringe settings: `FR 0` (no fringe), `FR 3` (third-order fringe), and `FR 3` with `FC` (Enge coefficients, where available).

Table 3: Vertical edge kick M_{43} and Y-plane stability for representative dipole triplets. FELsim uses a triangle-model fringe correction; COSY uses the `DIL` element under different `FR` settings.

Dipole	θ (°)	FELsim	COSY FR 0	COSY FR 3	COSY FR 3+FC
M_{43} *(vertical kick,* rad/m*)*					
Transport	1.5	-7.42×10^{-3}	-7.71×10^{-3}	-6.78×10^{-3}	—
Chicane	4.0	-1.11×10^{-1}	-1.21×10^{-1}	-8.93×10^{-2}	—
FC1	11.25	-9.50×10^{-1}	-1.04	-7.79×10^{-1}	-2.27×10^{-1}
$\lvert\mathrm{Tr}(M_y)/2\rvert$ *(stability, < 1 required)*					
Transport	1.5	0.9997	0.9997	0.9997	—
Chicane	4.0	0.9977	0.9975	0.9982	—
FC1	11.25	0.9823	0.9805	0.9855	0.9957

Three findings emerge from Table 3:

1. FELsim's Y-kick lies between COSY FR 0 and FR 3 for every dipole type, but closer to FR 0 (the hard-edge model).

2. The discrepancy grows with bending angle: 3.9% at 1.5°, 8.9% at 4°, and 10% at 11.25° (FR 0 vs FELsim).

3. The `FC` Enge coefficients (available for the FC1/FC2 chicane dipoles) produce much weaker Y-kicks (76% less than FELsim), indicating that realistic fringe fields partially cancel the geometric edge focusing.

The cumulative effect of all 14 dipole sandwiches in FELsim gives $|\mathrm{Tr}(M_y)/2| = 1.203$ — formally *unstable* without quadrupoles. The quads must therefore provide sufficient Y-plane focusing to stabilise the beamline. When FELsim-optimised currents (calibrated to FELsim's dipole model) are injected into COSY (with its different edge kicks), the compensation breaks down, explaining the Y-plane instability observed in cross-validation ($|\mathrm{Tr}(M_y)/2| = 29$).

3.6 COSY FIT with First-Order Fringe Fields (FR 1)

To assess the impact of dipole fringe fields on the optimisation, we re-ran the full 11-stage COSY FIT with the `FR 1` command enabled, which includes first-order fringe field effects (Enge-model based Runge-Kutta integration through the fringe region).

Cold-start failure. Running FR 1 from the same default initial guesses as FR 0 produces poor convergence: MSE $= 1.7 \times 10^{-1}$ at order 3 and 5.7×10^{-1} at order 1. The Y-plane becomes unstable ($|\mathrm{Tr}(M_y)/2| = 1.43$) because the changed dipole edge kicks push the sequential FIT stages into incompatible local minima. The gradient-based FIT (algorithm 1) cannot recover once early stages converge to a wrong basin.

Warm-start from FR 0. Using the converged FR 0 quadrupole currents as initial guesses for FR 1 resolves the convergence problem completely. The optimiser needs only small adjustments from the nearby FR 0 solution:

Run	MSE	$\Delta I_{\max}$ (A)
FR 0 (baseline)	4.50×10^{-9}	—
FR 1 (cold start)	1.66×10^{-1}	—
FR 1 (warm from FR 0)	7.93×10^{-8}	0.76

The warm-started FR 1 run achieves MSE $= 7.9 \times 10^{-8}$, comparable to the FR 0 baseline, with a maximum current change of 0.76 A. Stages 1–2 and 6–9 barely change; the main adjustments are in the triplets (Stages 3, 5, 10–11).

Table 4 compares the quadrupole currents across all three converged runs: FELsim, COSY FR 0, and COSY FR 1.

FR 3 computational cost. Third-order fringe field computation (`FR 3`) was attempted but is prohibitively expensive: each lattice evaluation takes >15 minutes due to Runge-Kutta integration through 14 dipole fringe regions, even at first-order DA. A full 11-stage FIT (requiring hundreds of lattice evaluations) would take days. The single-dipole comparison (Table 3) provides FR 3 results for individual elements.

Table 4: Optimised quadrupole currents: FELsim, COSY FR 0, and COSY FR 1 (warm-started from FR 0).

Elem	Section	FELsim (A)	COSY FR 0 (A)	COSY FR 1 (A)	Stage
1	First doublet	0.822	0.839	0.842	1
3	First doublet	1.043	1.052	1.057	1
10	Chromaticity 1	3.883	3.957	3.979	2
16	Triplet 1	2.240	1.993	1.994	3
18	Triplet 1	4.953	4.859	4.955	3
20	Triplet 1	3.426	3.499	3.599	3
27	Chromaticity 2	4.666	4.685	4.711	4
33	Double triplet	2.694	0.357	0.518	5
35	Double triplet	2.652	1.975	1.849	5
37	Double triplet	0.277	1.995	1.985	5
50	Chromaticity 3	4.674	4.685	4.696	6
56	IP doublet	3.122	3.168	3.218	7
58	IP doublet	3.313	3.321	3.361	7
61	Post-IP doublet	5.178	5.166	5.252	8
63	Post-IP doublet	4.043	4.011	4.080	8
70	Chromaticity 4	4.682	4.662	4.811	9
76	Triplet 2	3.934	3.869	3.953	10
78	Triplet 2	4.079	4.452	4.506	10
80	Triplet 2	0.014	0.608	0.592	10
87	Chromaticity 5	1.362	1.259	1.610	11
93	Final triplet	0.945	2.348	2.473	11
95	Final triplet	2.885	3.802	3.602	11
97	Final triplet	2.192	4.747	3.995	11

4 Physics Model Differences

4.1 Dipole Edge Fringe Fields

FELsim's `dipole_wedge` class applies a simplified triangle-model fringe field correction to the vertical edge kick:

$$T_y = \tan\left(\epsilon - \frac{\ell_e}{6}\,\frac{1+\sin^2\epsilon}{\rho\cos\epsilon}\right), \tag{2}$$

where ℓ_e is the physical DPW length, ϵ is the edge angle, and ρ is the bending radius. COSY's `DIL` with $\beta = 0$ (our FR 0 baseline) gives $T_y = \tan\epsilon$ without fringe correction, while FR 3 computes fringe fields to third order using the half-gap parameter.

Each individual dipole shows a 4–20% difference in the Y-kick (Table 3). Over 14 dipoles (28 edge kicks), this accumulates to Y-plane instability when currents are exchanged between codes.

COSY provides more physically accurate fringe field treatments: the `FC` command with Enge function coefficients (measured values are available for the FC1/FC2 dipoles), the `FR` command for higher-order fringe fields, and the `MGE` command for 1D field map integration. The `MGE` approach requires a correctly scaled field map (currently unavailable for this beamline).

4.2 Current Sign Handling

FELsim's quadrupole model uses $k = |eGI/(m_e c\beta\gamma)|$, making the transfer matrix independent of the sign of I. COSY's `MQ` uses $B_{\text{pole}} = \pm GIr$, where the sign determines the focusing plane. Consequently, COSY's FIT can explore negative-current solutions that reverse the focusing/defocusing planes, while FELsim's Nelder-Mead (with bounds $[0, 10]$ A) is restricted to the positive quadrant.

This explains why COSY consistently finds the negative-polarity Stage 5 solution: it is a genuine local minimum accessible to COSY but not to FELsim's bounded optimiser. Both solutions satisfy the Stage 5 objectives.

5 Summary

1. COSY INFINITY FIT successfully reproduces the 11-stage beamline optimisation in a single execution, achieving MSE $\sim 10^{-8}$–10^{-9} for both FR 0 and FR 1 (warm-started). FELsim's Nelder-Mead achieves MSE $= 2.9 \times 10^{-5}$.
2. All three codes/settings match the undulator Twiss targets to high precision. The lower COSY MSE reflects the advantage of gradient-based (DA) optimisation over Nelder-Mead.
3. Enabling first-order fringe fields (`FR 1`) requires warm-starting from the FR 0 solution. Cold-start with FR 1 fails (MSE ~ 0.2) because the changed dipole edge kicks push the sequential FIT stages into incompatible local minima.
4. Third-order fringe fields (`FR 3`) are prohibitively expensive: each lattice evaluation takes >15 minutes due to Runge-Kutta integration through 14 dipole fringe regions. Single-dipole comparisons (Table 3) quantify the FR 3 effect.
5. The principal discrepancy is Stage 5, where COSY converges to negative-polarity currents while FELsim (constrained to $[0, 10]$ A) finds all-positive currents (Section 4).
6. Initial Twiss from the beam distribution ($\beta_0 = \sigma^2/\varepsilon \approx 6.34\,\text{m}$) is critical for correct transport computation.
7. Direct current injection between codes fails due to the dipole edge fringe correction difference (5–8% per edge, accumulating to Y-plane instability). The codes agree on final Twiss when each optimises within its own model.
8. S1 and S3 produce identical transverse optimisation results.

Next steps.

- Obtain the correct scaling coefficient for the chicane dipole field map to enable MGE-based simulation (the most physically accurate dipole model available in COSY).
- Add η'_x or $\eta_x(117)$ objective to control undulator dispersion.
- Constrain Stage 5 variables to $I > 0$ (e.g., via I^2 parameterisation) to find the positive-polarity COSY solution.

References

[1] M. Weinberg, A. Fisher, and R. Li, "Design of the UH Mark V Free Electron Laser," arXiv:2510.14061v1, Table I.